Lesson 1: Right-Triangle Trigonometry: Sine, Cosine, and Tangent

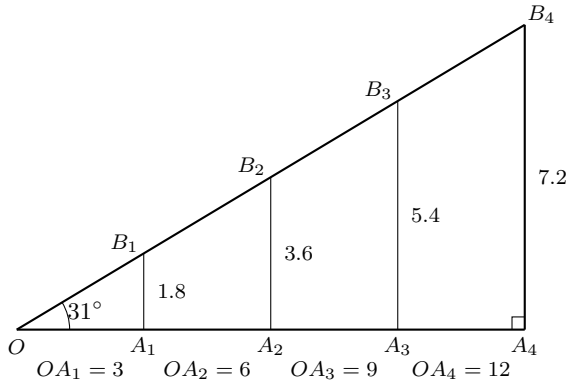
Fix one acute angle of a right triangle and every ratio of its sides is fixed — those ratios are sine, cosine and tangent.

1. Similar Right Triangles and a Constant Ratio

Two right triangles that have an *equal* acute angle are similar, so the ratio of any pair of corresponding sides is the same in both — no matter how large or small each triangle is.

Example 1 — A ratio that depends only on the angle

The four right triangles $\triangle OA_1B_1$, $\triangle OA_2B_2$, $\triangle OA_3B_3$ and $\triangle OA_4B_4$ all share the angle of $31°$ at O. Using the side lengths shown, compute each ratio $\dfrac{A_iB_i}{OA_i}$ and round it to one decimal place. The first row is done for you.



$$\frac{A_1B_1}{OA_1} = \frac{1.8}{3} = 0.6$$

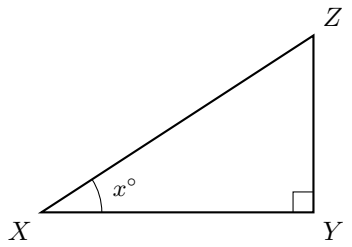
$$\frac{A_2B_2}{OA_2} = \frac{3.6}{6} = \underline{\qquad\qquad}$$

$$\frac{A_3B_3}{OA_3} = \frac{5.4}{9} = \underline{\qquad\qquad}$$

$$\frac{A_4B_4}{OA_4} = \frac{7.2}{12} = \underline{\qquad\qquad}$$

The ratio _____ is therefore connected to the angle of $31°$ itself. Evaluate $\tan(31°)$ on a calculator, to the nearest thousandth, and compare it to the ratios above: _____

2. Naming the Sides of a Right Triangle



In right triangle XYZ, with the right angle at Y and $\angle YXZ = x°$, each side is named *relative to* $x°$:

- XZ, the side opposite the right angle, is the **hypotenuse** (hyp) — always the longest side, and always the same side whichever acute angle is used.
- YZ, the side opposite $x°$, is the **opposite** side (opp).
- XY, which forms $x°$ together with the hypotenuse, is the **adjacent** side (adj).

Using $\angle XZY$ as the reference angle instead swaps opp and adj: the hypotenuse never changes, but "opposite" and "adjacent" are always defined relative to the angle being used.

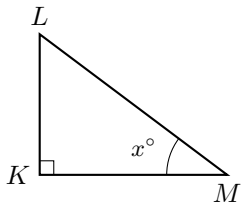
Labelling routine (use it every time). **1.** Label the **hyp** first — the side across from the right angle. **2.** The **opp** side does *not* touch the reference angle. **3.** The **adj** side *touches* the reference angle (and is not the hypotenuse).

3. Identifying Sides for a Given Reference Angle

Example 2 — Reading opp, adj and hyp off a figure

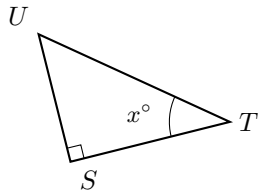
For the reference angle x° marked on each triangle, label the hypotenuse (hyp), the opposite side (opp), and the adjacent side (adj).

(a)



hyp = _____ opp = _____
adj = _____

(b)



hyp = _____ opp = _____
adj = _____

4. SOH CAH TOA

For a reference angle x° in a right triangle, the three **primary trigonometric ratios** are

$$\sin x^\circ = \frac{\text{opp}}{\text{hyp}} \qquad \cos x^\circ = \frac{\text{adj}}{\text{hyp}} \qquad \tan x^\circ = \frac{\text{opp}}{\text{adj}}$$

memorized as **SOH CAH TOA**: Sine is Opposite over Hypotenuse; Cosine is Adjacent over Hypotenuse; Tangent is Opposite over Adjacent. The ratio found in Example 1 was exactly the tangent ratio, $\frac{\text{opp}}{\text{adj}}$.

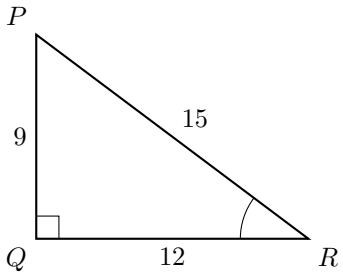
Size check. The hypotenuse is always the longest side, so for an acute angle $\sin x^\circ$ and $\cos x^\circ$ are always **less than 1**. The tangent compares the two legs, so it can be any positive number: less than 1 when opp $<$ adj, and greater than 1 when opp $>$ adj.

5. Computing the Ratios From Given Side Lengths

When all three side lengths of a right triangle are known, each ratio for either acute angle can be written down directly from SOH CAH TOA.

Example 3 — The same triangle, the two different reference angles

Right triangle PQR is shown, with the right angle at Q.

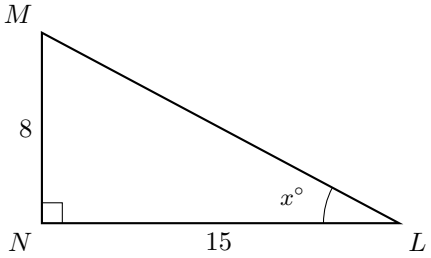


(a) Using $\angle R$ as the reference angle, write $\sin R$, $\cos R$ and $\tan R$ as fractions in simplest form.

(b) Now use $\angle P$ as the reference angle. Write $\sin P$, $\cos P$ and $\tan P$ as fractions in simplest form. How do these compare to your answers in (a)?

Example 4 — Finding the missing hypotenuse first

In right triangle LMN the right angle is at N, $NM = 8$ and $NL = 15$. The hypotenuse LM is not given.



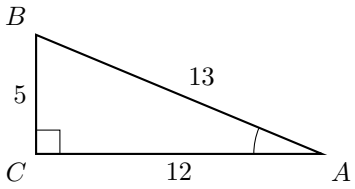
(a) Use the Pythagorean theorem to determine the length of LM.

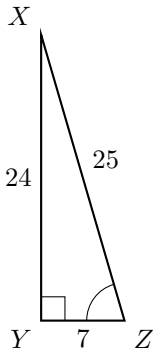
(b) Using $\angle L = x^\circ$ as the reference angle, write $\sin x^\circ$, $\cos x^\circ$ and $\tan x^\circ$ as fractions in simplest form, and then as decimals rounded to the nearest thousandth.

6. More Practice With Two Triangles

Example 5 — Two triangles, two different reference angles

Two right triangles are shown below.





Determine each ratio as a fraction in simplest form.

(i) $\sin A =$ _____ (ii) $\cos A =$ _____ (iii) $\tan A =$ _____

(iv) $\sin Z =$ _____ (v) $\cos Z =$ _____ (vi) $\tan Z =$ _____

Lesson 2: Trig Ratios: Calculator Values and Angle Patterns

Evaluating the ratios in degree mode, and how each ratio behaves as the angle opens from 0° to 90°. (Finding an angle from a ratio comes in Lesson 4.)

1. Evaluating Trigonometric Ratios with a Calculator

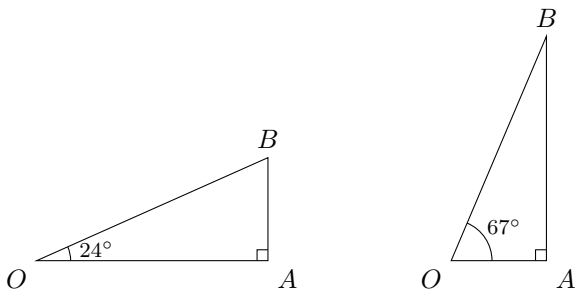
For a reference angle x° in a right triangle, SOH CAH TOA gives

$$\sin x^\circ = \frac{\text{opp}}{\text{hyp}} \qquad \cos x^\circ = \frac{\text{adj}}{\text{hyp}} \qquad \tan x^\circ = \frac{\text{opp}}{\text{adj}}$$

A calculator returns any of these ratios, for any angle between 0° and 90°, to as many decimal places as are wanted — but only once it is set to **degree mode**.

Example 1 — Calculator values, and the side ratio each one names

Each triangle below has a right angle at A.



(a) Use a calculator to determine the value of each ratio, to four decimal places.
(i) $\sin 24° =$ (ii) $\cos 24° =$ (iii) $\sin 67° =$ (iv) $\cos 67° =$

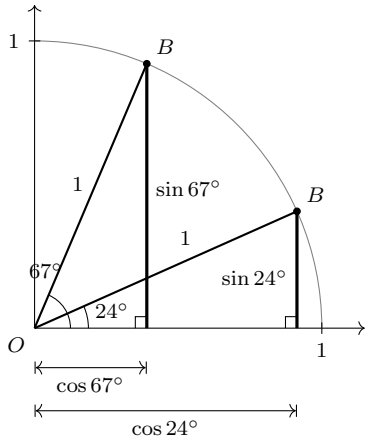
(b) With respect to the two diagrams above, which ratio of sides ($\frac{OA}{OB}$, $\frac{AB}{OB}$, $\frac{AB}{OA}$, etc.) is represented by each of the four ratios in (a)?

2. How the Ratios Change as the Angle Changes

Rescaling a right triangle so that its hypotenuse is exactly 1 unit long makes the sine and cosine ratios equal to the side lengths themselves: $\sin x° = \dfrac{\text{opp}}{1} = \text{opp}$ and $\cos x° = \dfrac{\text{adj}}{1} = \text{adj}$. Drawing every such triangle inside one quarter circle of radius 1 lets the effect of opening the angle be read straight off the picture.

Example 2 — Why sine grows and cosine shrinks as the angle opens

The diagram shows the two triangles of Example 1 rescaled so that each hypotenuse OB equals 1 unit. Both hypotenuses are radii of the same quarter circle, so the **vertical** legs (the sines) and the **horizontal** legs (the cosines) can be compared directly.



(a) As angle AOB increases from $24°$ to $67°$, explain — using SOH CAH TOA and the diagrams above — why the sine ratio *increases* in value while the cosine ratio *decreases* in value.

(b) Determine, with reasoning, what happens to the tangent ratio as the angle increases from $24°$ to $67°$.

3. Comparing Ratios Without a Calculator

Once it is clear *why* each ratio grows or shrinks with the angle, two ratios can be compared by reasoning alone.

The $45°$ benchmark. At exactly $45°$ the triangle is isosceles: opp = adj. So $\sin 45° = \cos 45°$ and $\tan 45° = 1$. Below $45°$ the adjacent leg is longer ($\cos > \sin$, $\tan < 1$); above $45°$ the opposite leg is longer ($\sin > \cos$, $\tan > 1$).

Example 3 — Ordering ratios by SOH CAH TOA reasoning

Without using a calculator, state which ratio in each part has the greater value. Justify each answer using SOH CAH TOA reasoning.

(a) $\sin 18°,\ \sin 71°$

(b) $\cos 12°,\ \cos 55°$

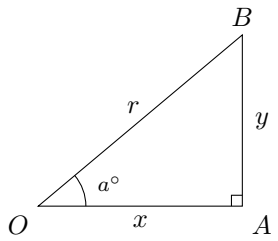
(c) $\tan 22°,\ \tan 80°$

(d) $\sin 50°,\ \cos 50°,\ \tan 50°$

4. Special Ratios at $0°$ and $90°$, and the Pattern Across All Angles

A triangle cannot actually be formed when the angle is $0°$ or $90°$, but the ratios still make sense if a rotation angle is pictured sweeping open. Below, a segment of length r is rotated counterclockwise about O through an angle of $a°$, forming right triangle OAB with opposite side y, adjacent side x, and hypotenuse r.

Example 4 — The two extremes, and the pattern in between



Given that $\sin a° = \dfrac{y}{r}$, **(a)** write $\cos a°$ and $\tan a°$ in terms of x, y, and r.

(b) Explain what happens to x and y as a shrinks toward $0°$, and what happens to x and y as a grows toward $90°$. Use this to state the exact value of each of $\sin 0°$, $\cos 0°$, $\tan 0°$, $\sin 90°$, $\cos 90°$, and $\tan 90°$.

(c) Complete the table below, rounding to two decimal places where necessary, then complete the statement underneath it.

Angle $x°$	$0°$	$15°$	$30°$	$45°$	$60°$	$75°$	$90°$
$\sin x°$							
$\cos x°$							

As $x°$ increases from $0°$ to $90°$, $\sin x°$ _____ from _____ to _____, while $\cos x°$ _____ from _____ to _____.

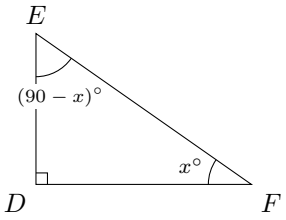
(d) *(Extension)* Use a calculator to complete: $\tan 89° =$ _____, $\tan 89.9° =$ _____, $\tan 89.99° =$ _____. Then complete the statement: as $x°$ increases from $0°$ toward $90°$, $\tan x°$ _____.

5. Extension — Complementary Angles: the Cofunction Relationship

In any right triangle the two acute angles add to $90°$ — they are **complementary**. This links the sine of one acute angle to the cosine of the other.

Example 5 — $\sin x° = \cos(90 - x)°$

Triangle DEF has a right angle at D, with $\angle F = x°$, as shown.



(a) Write an expression for $\angle E$ in terms of x.

(b) Explain, using the side ratios in the diagram, why $\sin x° = \cos(90 - x)°$.

(c) Complete the following, using an angle between $0°$ and $90°$.

(i) $\sin 63° = \cos$ _____ (ii) $\sin 9° = \cos$ _____

Lesson 3: Finding a Missing Side Length in a Right Triangle

Name the sides relative to the given angle, choose the ratio that holds the known side and the unknown, then multiply or divide.

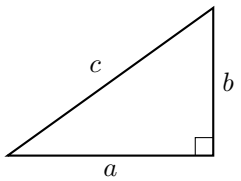
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1. Review: The Pythagorean Theorem

In a right triangle with legs a and b and hypotenuse c,

$$c^2 = a^2 + b^2, \qquad a^2 = c^2 - b^2, \qquad b^2 = c^2 - a^2.$$

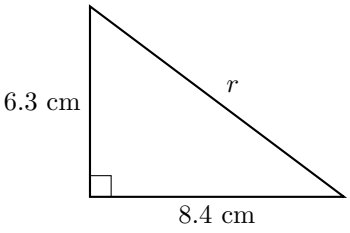
The theorem applies whenever *two* side lengths are known. When one side length and one acute angle are known instead, a trigonometric ratio is needed — but recognizing which tool the given information calls for is the first decision in every question.



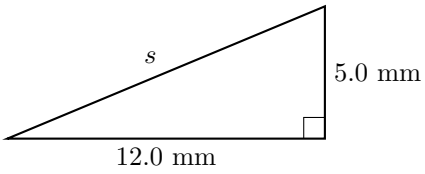
Example 1 — two sides known, so no trigonometry is required

Calculate the length of the third side of each triangle, to the nearest tenth of a unit if necessary.

(a)



(b)



2. The Unknown in the Numerator

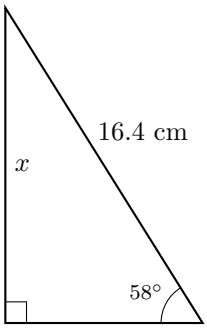
Once an acute angle is chosen, every side of the triangle has a name relative to it: the **opposite** side, the **adjacent** side, and the **hypotenuse**. SOH CAH TOA then fixes the ratio to use:

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}, \qquad \cos\theta = \frac{\text{adj}}{\text{hyp}}, \qquad \tan\theta = \frac{\text{opp}}{\text{adj}}.$$

If the unknown side is on top of the chosen ratio, the equation is solved by a single multiplication: $\dfrac{x}{k} = \sin\theta$ gives $x = k\sin\theta$.

Example 2 — setting up the ratio with the unknown on top

The measure of an acute angle and the length of the hypotenuse are given. Complete the steps for finding x, the side opposite the given angle.



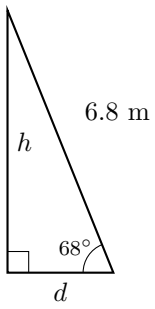
The given side, 16.4 cm, is the _____, and the side to be found, x, is _____ the given angle of 58°.
We use the _____ ratio.

$$\frac{x}{16.4} = _____58^\circ \qquad x = 16.4\ _____58^\circ$$

To the nearest tenth, $x =$ _____ cm.

Example 3 — a ladder against a wall

A ladder 6.8 m long leans against a wall, making an angle of 68° with the ground.



(a) How high up the wall, to the nearest tenth of a metre, does the ladder reach?

(b) Use trigonometry to determine, to the nearest tenth of a metre, the distance d between the foot of the ladder and the base of the wall.

(c) Use your answer to (a) and the Pythagorean theorem to determine d again, to the nearest tenth of a metre.

(d) Explain why the answers to (b) and (c) are not quite the same.

3. The Unknown in the Denominator

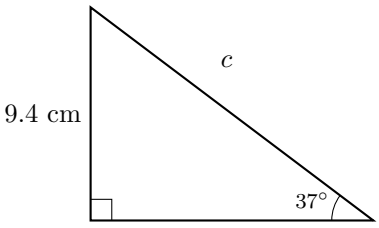
If the unknown side is the hypotenuse — or, more generally, sits underneath the known side in the ratio (for example the *adjacent* leg in $\tan\theta = \frac{\text{opp}}{\text{adj}}$ when the opposite leg is the one given) — it lands in the *denominator*, and the equation is solved by cross-multiplying and then dividing:

$$\sin\theta = \frac{k}{x} \quad \Rightarrow \quad x\sin\theta = k \quad \Rightarrow \quad x = \frac{k}{\sin\theta}, \qquad \text{and likewise} \qquad \tan\theta = \frac{k}{x} \quad \Rightarrow \quad x = \frac{k}{\tan\theta}.$$

Where the unknown sits decides the whole calculation: on top it is a multiplication, underneath it is a division. Carry full calculator precision through and round only the final answer.

Example 4 — cross-multiplying to free the unknown

The acute angle and the leg opposite it are known, but the hypotenuse c is unknown, so c appears in the <u>denominator</u> of the ratio. Complete the steps.



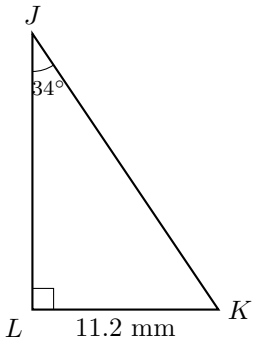
$$_____37^\circ = \frac{9.4}{c}$$ cross multiply $$c\ _____37^\circ = 9.4$$

Divide both sides by _____37° to get

$$c = \frac{9.4}{_____37^\circ} = _____ \text{ cm (to the nearest tenth).}$$

Example 5 — reading the labels from a vertex-letter description

Determine, to one decimal place, the length of the hypotenuse JK of triangle JKL, in which angle $JLK = 90°$, angle $LJK = 34°$, and $KL = 11.2$ mm.

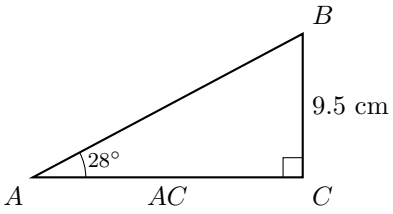


4. Solving for More Than One Side

A single triangle may ask for two unknown sides at once — typically one numerator case and one denominator case. Set each one up from the *given* side and the *given* angle; never build the second calculation on the rounded answer to the first.

Example 6 — an unknown leg in the denominator of tangent

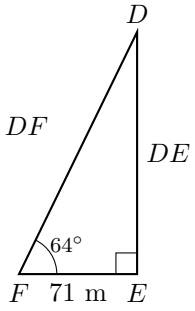
In right triangle ABC, $\angle C = 90°$, $\angle A = 28°$, and the leg opposite $\angle A$ is $BC = 9.5$ cm. Determine the length of the adjacent leg AC, to the nearest tenth of a centimetre.



Example 7 — one triangle, two unknown sides

In triangle DEF, angle $DEF = 90°$, angle $DFE = 64°$, and $EF = 71$ m. Use trigonometric ratios to determine the lengths of DE and DF, each to the nearest metre.

(a) Determine the length of DE.



(b) Determine the length of DF.

5. Summary — Which Tool? Where Is the Unknown?

What is known	Tool	Where is the unknown? What do I do?
two sides, no acute angle	Pythagorean theorem	$c^2 = a^2 + b^2$; subtract first if the unknown is a leg
an acute angle and one side	SOH, CAH or TOA — the ratio that holds the known side *and* the unknown	unknown on **top**: **multiply**, e.g. $x = k \sin\theta$ unknown **underneath**: **divide**, e.g. $x = k \div \sin\theta$ or $x = k \div \tan\theta$
only angles, or no right angle	—	not enough information for these tools

Sense check every answer: the hypotenuse is the longest side, and the leg opposite the larger acute angle is the longer leg.

Lesson 4: Finding the Measure of an Angle in a Right Triangle

Two sides known and the angle unknown — name the sides, choose the matching ratio, then apply $\sin^{-1}$, $\cos^{-1}$ *or* $\tan^{-1}$.

1. Inverse Trigonometric Ratios

A calculator can also run SOH CAH TOA *backward*. If the value of a trigonometric ratio for an acute angle is known, the angle itself is recovered with the **inverse ratio**: if $\sin\theta^\circ = k$ then $\theta^\circ = \sin^{-1}(k)$, and likewise $\cos^{-1}$ and $\tan^{-1}$. On most calculators these are 2nd sin, 2nd cos and 2nd tan — in **degree mode**. Read $\sin^{-1}(k)$ as "the angle whose sine is k"; it does *not* mean $1 \div \sin$.

Example 1 — Ratio to value, and value back to angle

(a) Use a calculator to determine the value of each ratio, to four decimal places.

(i) $\sin 79^\circ =$ (ii) $\tan 11^\circ =$ (iii) $\cos 52^\circ =$

(b) Now run it backward: in each case, determine the measure of the acute angle to the nearest degree.

(i) $\sin a^\circ = 0.4217$, $a^\circ =$ (ii) $\cos b^\circ = 0.6820$, $b^\circ =$ (iii) $\tan P = 2.7500$, $\angle P =$ (iv) $\sin C = \frac{3}{5}$, $\angle C =$

2. Determining an Angle From Two Known Sides

Calculating an angle requires the lengths of **two** of the triangle's three sides. Name each known side relative to the unknown angle — opposite, adjacent or hypotenuse — choose the ratio that pairs those two names (SOH, CAH or TOA), write it as an equation, then solve with the inverse ratio. Reading the ratio off the wrong pair of sides is the usual error: name the sides first, choose the ratio second.

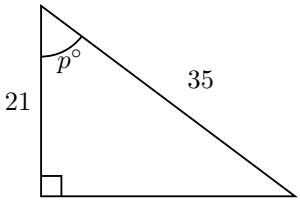
Naming the sides before choosing the ratio

Complete the following for the diagram shown.

Relative to the angle p°, 21 is the length of the ADJACENT side and 35 is the length of the _____.

The trigonometric ratio which involves the ADJACENT side and the _____ is the _____ ratio.

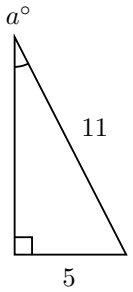
_____ $p^\circ = \frac{21}{35}$ so $p^\circ =$ _____



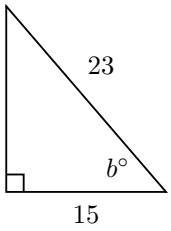
Example 2 — Two sides given, angle to the nearest degree

In each case, determine the measure of the indicated angle to the nearest degree.

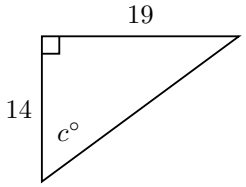
(a)



(b)

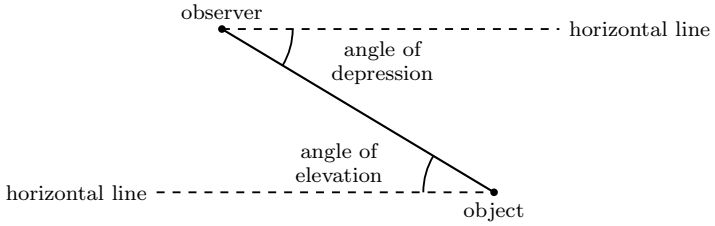


(c)



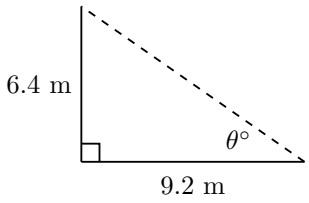
3. Angle of Elevation and Angle of Depression

The **angle of elevation** is measured **upward** from a horizontal line to the line of sight of an object above the horizontal; the **angle of depression** is measured **downward** from a horizontal line to the line of sight of an object below it. Both are measured from the **horizontal**, never from the vertical. The horizontal line at the observer and the horizontal line at the object are parallel, and the line of sight is a transversal crossing both, so the alternate interior angles they form are equal: **the angle of depression from the higher point = the angle of elevation from the lower point.** In the diagram, the angle of depression of the object from the observer equals the angle of elevation of the observer from the object.



Example 3 — Angle of elevation of the sun

A flagpole 6.4 m tall casts a shadow 9.2 m long. Determine, to the nearest tenth of a degree, the angle of elevation of the sun.



Example 4 — Angle of depression from the top of a cliff

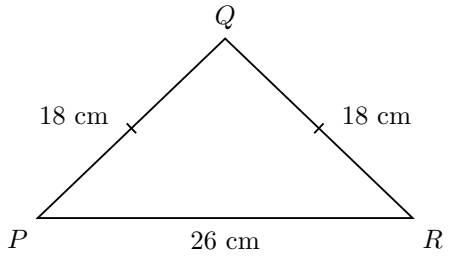
A sea kayaker is on the water below a cliff. The cliff is 68 m high, and the kayaker is 155 m from the base of the cliff.
(a) Draw a diagram to represent this situation, and mark the angle of depression of the kayaker from the top of the cliff.
(b) Determine, to the nearest degree, the angle of depression of the kayaker from the top of the cliff.

4. Angles in Isosceles Triangles

SOH CAH TOA applies only inside a **right** triangle. An isosceles triangle is not in general a right triangle, but dropping a perpendicular from its apex to its base splits it into two **congruent right triangles**, each of which the ratios do apply to. The perpendicular bisects both the base and the apex angle, so the angle found in one half is *half* of the apex angle.

Example 5 — Splitting an isosceles triangle

Consider isosceles triangle PQR, in which $PQ = 18$ cm, $QR = 18$ cm, and $PR = 26$ cm.



(a) Why can SOH CAH TOA not be used directly in triangle PQR to determine the measure of angle PQR?

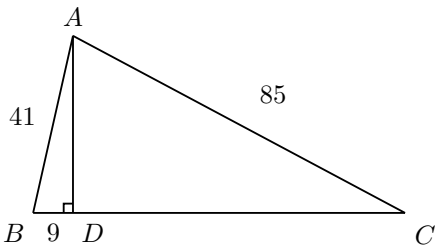
(b) Determine, to the nearest degree, the measure of angle PQR by splitting triangle PQR into two congruent right triangles.

5. Combining Two Right Triangles

Some diagrams show two right triangles joined along a common side. The Pythagorean Theorem gives the length of the shared side, and the two triangles are then used together — one angle from each — to find an angle that spans both.

Example 6 — An angle split across two right triangles

The diagram shows two right triangles sharing the common side AD, where D lies on BC between B and C.



(a) Use the Pythagorean Theorem to calculate the length of AD.

(b) Determine, to the nearest degree, the measure of angle BAC.

Lesson 5: Solving Right Triangles Completely

Every side and every angle — take each unknown from the original given data and round only at the end.

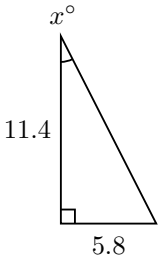
1. Choosing the Right Tool

Two known measurements determine a third. Two sides with no angle involved call for the **Pythagorean Theorem**; a side paired with an acute angle, or two sides with the angle unknown, call for **SOH**, **CAH** or **TOA**. Name the given sides relative to the angle in use — opposite, adjacent, hypotenuse — then pick the tool that uses exactly what is given.

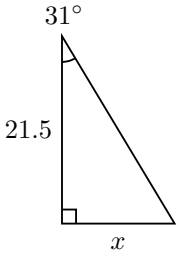
Example 1 — One unknown: match the tool to the given pair

In each case, calculate the indicated measure. Give angles to the nearest degree and sides to the nearest tenth.

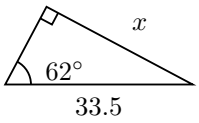
(a)



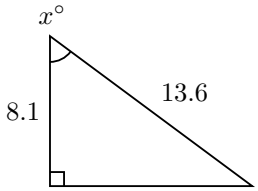
(b)



(c)



(d)

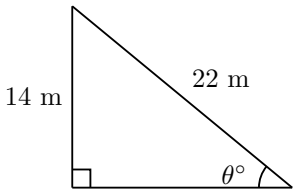


2. Two Routes to the Same Side

When a side can be reached either by the Pythagorean Theorem or by a trigonometric ratio, the two routes agree only if no rounded value is fed into them. The next example computes one length both ways and compares.

Example 2 — The same length, two ways

A radio antenna is 14 m tall. It is supported by a guy wire 22 m long running from the top of the antenna to a stake in the ground.



(a) Calculate, to the nearest degree, the angle $\theta°$ between the wire and the ground.

(b) Calculate the distance, to the nearest tenth of a metre, between the stake and the foot of the antenna, using

i) the Pythagorean Theorem

ii) trigonometry, using your rounded angle from (a).

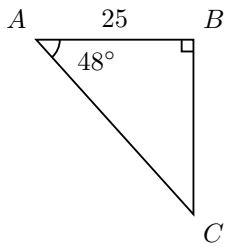
3. What It Means to "Solve" a Triangle

Solving a right triangle means finding the length of **every** side and the measure of **every** angle. One angle — the right angle — is already known, so exactly **two** more independent pieces of information are needed: either one side and one acute angle, or two sides. From those, the Pythagorean Theorem, SOH CAH TOA, and the fact that the two acute angles of a right triangle add to 90° deliver everything else, applying whichever tool fits the information available at each step.

Take every unknown from the **original given measurements** wherever possible, not from a value you have just rounded. A rounded intermediate result carries its error into everything built on it, as Example 2 shows. Carry full calculator precision through the work and round only the final answer.

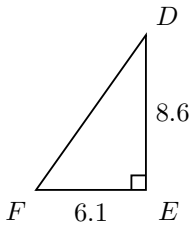
Example 3 — Solve from one angle and one side

Solve the given triangle. Give all angles to the nearest degree and all sides to the nearest tenth.



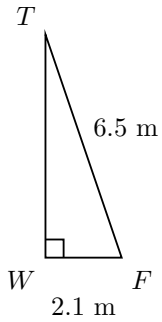
Example 4 — Solve from two sides

Solve triangle DEF. Give all sides to the nearest tenth and all angles to the nearest degree.



Example 5 — Solve an applied triangle

A 6.5 m ladder leans against a vertical wall with its foot 2.1 m from the base of the wall. Let W be the point where the wall meets the ground, F the foot of the ladder, and T the point where the top of the ladder touches the wall. Solve triangle WFT: find the height WT the ladder reaches on the wall, the angle the ladder makes with the ground at F, and the angle the ladder makes with the wall at T. Give the side to the nearest tenth of a metre and both angles to the nearest degree.



Lesson 6: Multi-Step Trigonometry: Eye Level, Elevation and Depression

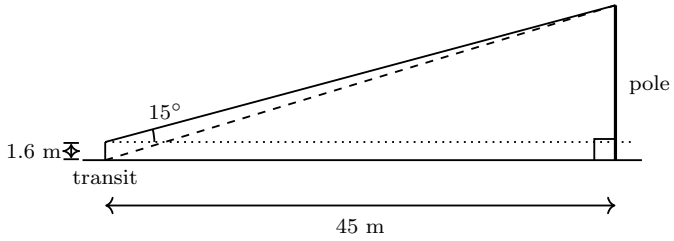
Layered right-triangle problems — eye-level heights, hidden right angles, and indirect measurement with a clinometer. (Extension: tangent circles and bearings.)

1. Multi-Step Problems with a Single Triangle

Some situations are solved inside a single right triangle but still need two separate calculations — one for part (a), and a second for part (b) that uses a different vertex or a different pair of sides. Read every part of a multi-part problem before starting: a later part frequently reuses a length found in an earlier one.

Example 1 — Two calculations in one triangle

Land surveyor Priya uses a transit (a tripod-mounted sighting instrument) to determine the height of a utility pole. The transit is set up 45 m from the base of the pole, and the angle from the transit's eyepiece up to the top of the pole is measured to be 15°.



(a) If the transit's eyepiece is 1.6 m above the ground, calculate the height of the pole to the nearest tenth of a metre.

(b) Determine, to the nearest degree, the angle of elevation of the top of the pole from the *base* of the transit (at ground level, not the eyepiece).

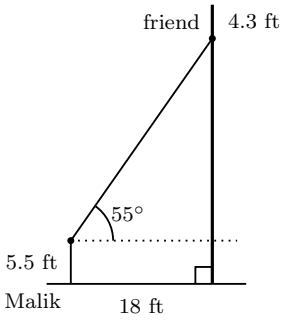
Hidden right angles. SOH CAH TOA needs a right angle, and sometimes the right angle is not marked. One you met in Math 9: an angle **inscribed in a semicircle** — its vertex on the circle and its two arms through the ends of a diameter — is always 90°.

2. Angles Measured from Eye Level

An angle of elevation or depression is often measured from an observer's **eye level**, not from the ground. Before applying SOH CAH TOA, check whether the diagram's vertical measurements need to be adjusted by an eye height: the trigonometry returns only the rise *above eye level*, so the eye height must be added back at the end.

Example 2 — Eye level at both ends of the line of sight

Malik stands on the ground 18 ft from the base of an apartment building and looks up at his friend, who is leaning out of a window. The angle of elevation from Malik's eye level to his friend's eye level is 55°. Malik's eyes are 5.5 ft above the ground, and his friend's eyes are 4.3 ft below the very top of the building. Determine the height of the building to the nearest tenth of a foot.



3. Indirect Measurement With a Clinometer

A **clinometer** measures an angle of elevation. A simple one is a protractor with a drinking straw taped along its straight edge and a string, weighted with a washer, hanging from the centre hole. Sight the top of an object through the straw: the string hangs straight down, so the angle of elevation is the difference between the string's reading and 90°:

$$\text{angle of elevation} = |\,90^\circ - \text{string reading}\,|\,.$$

Activity — How tall is it? Work in groups of three (sighter, reader, recorder).

1. Build the clinometer: tape the straw along the protractor's straight edge; tie the string through the centre hole and hang a washer on it.
2. Measure the sighter's **eye height** with a metre stick.
3. Measure the **horizontal distance** from the sighter's feet to the base of a tall object (flagpole, wall, goalpost) with a measuring tape or metre stick.
4. Sight the top of the object through the straw. The reader records where the string crosses the protractor; convert it to the angle of elevation.
5. Calculate: height = eye height + distance $\times \tan$(angle of elevation).
6. Measure the height **directly** where you can (a wall or doorframe with the metre stick, or count bricks $\times$ the height of one brick) and compare.

Object	eye height (m)	distance (m)	string reading	angle of elevation	calculated height (m)	direct height (m)

Example 3 — Reading a clinometer

Ana's eyes are 1.55 m above the ground. Standing 18.0 m from the base of the school flagpole, she sights the top through her clinometer, and the string crosses the protractor at 58°.

(a) What is the angle of elevation?

(b) Determine the height of the flagpole to the nearest tenth of a metre.

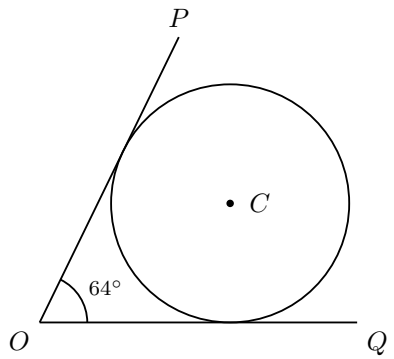
(c) The caretaker says the pole is 12.5 m tall. Give two reasons the measurement could be off.

4. Extension — Tangent Lines to a Circle

A line tangent to a circle meets the radius drawn to the point of tangency at a **right angle**. When two tangent segments are drawn to a circle from the same outside point, the two segments are equal in length, and the segment joining that outside point to the circle's centre **bisects** the angle between them. Both facts let a right triangle be built to work with, even though no right angle is drawn in the original picture.

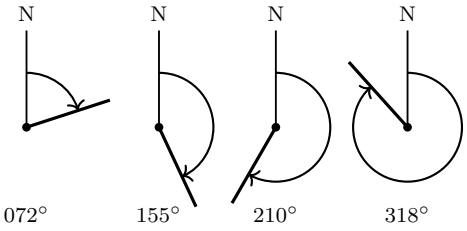
Example 4 — A circle tangent to both arms of an angle

A circular sprinkler head has centre C and radius 15 mm, and is tangent to both arms of angle POQ. If angle $POQ = 64^\circ$, determine the length of OC to the nearest millimetre.



5. Extension — Bearings

A **bearing** describes a direction as an angle measured **clockwise from north**, always written using three digits (for example 065°, not 65°). The four diagrams below show four different bearings, each measured at a single point; the ray shows the direction.



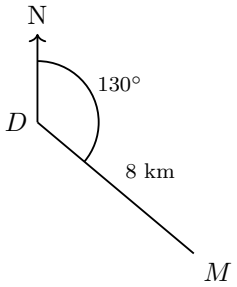
To use a bearing in a right triangle, work with the **acute angle between the direction and the north–south line** (its *reference angle*):

000°–090°	090°–180°	180°–270°	270°–360°
bearing, from north (toward E)	180° − bearing, from south (toward E)	bearing − 180°, from south (toward W)	360° − bearing, from north (toward W)

For a leg of length d, the leg is the hypotenuse: the east–west part is *opposite* the reference angle ($d\sin$) and the north–south part is *adjacent* ($d\cos$). The picture tells you which direction (N or S, E or W) each part goes. Draw a north–south line at every point where the direction of travel changes.

Example 5 — A ferry on a bearing

A ferry sails 8 km from a dock, D, on a bearing of 130° to a buoy, M, as shown.



(a) At M the ferry changes direction and sails due north until it reaches a point N that is due east of D. How far east of D is the ferry when it reaches N? Answer to the nearest 0.1 km.

(b) The ferry continues sailing north for a further 10 km to a point W. Determine the three-figure bearing of W from D.

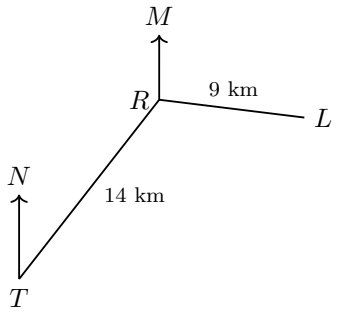
(c) If the ferry maintains an average speed of 12 km/h for the entire trip from D to W, determine, to the nearest tenth of an hour, the total time for the journey.

6. Extension — Combining Multiple Legs

When a journey has two or more legs on different bearings, draw a north–south line at *every* turning point. Split each leg of length d into an east–west part ($d\sin$ of its reference angle) and a north–south part ($d\cos$ of its reference angle), and use the picture to decide whether to add or subtract. The Pythagorean Theorem and an inverse tangent then give the direct distance and the bearing between the start and end points.

Example 6 — Two legs on different bearings

A cyclist leaves a trailhead, T, and rides 14 km on a bearing of 038° to a rest stop, R. At R she changes direction and rides 9 km on a bearing of 097° to a lookout, L. TN and RM represent north lines at T and R.



(a) On the diagram, mark the sizes of angle NTR and angle MRL.

(b) Determine, to the nearest 0.1 km, how far L is east of T.

(c) Determine, to the nearest 0.1 km, how far L is north of T.

(d) Determine, to the nearest 0.1 km, the direct distance from T to L.

(e) Determine the three-figure bearing of L from T.

Lesson 7: Chaining Right Triangles Together

Two right triangles joined by a shared side — solve the one that has enough information, then hand the shared side to the other.

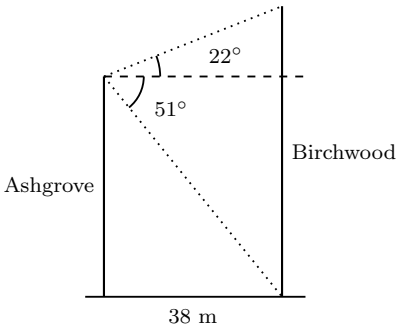
1. Indirect Measurement Across Two Structures

Two right triangles in a diagram are often joined along one side — an altitude, a vertical building, a common horizontal distance. That **shared side** is the link: solve the triangle in which enough is known, then carry the shared side into the second triangle to finish. **Never round the shared side.** Keep every value computed partway through a problem at full calculator accuracy and round only the final answer. Solve from the labelled numbers, never by measuring the picture.

When an observer at the top of one structure sights both the top and the base of a second structure, the angle of depression to the base gives the horizontal distance between the structures (using the first structure's height), and the angle of elevation to the top gives how much taller the second structure is. The two pieces add to the second structure's full height.

Example 1 — Two towers, one shared horizontal distance

Ashgrove Tower and Birchwood Tower stand 38 m apart. From the top of Ashgrove Tower, the angle of depression to the base of Birchwood Tower is 51°, and the angle of elevation to the top of Birchwood Tower is 22°. Determine the height of each tower, to the nearest metre.

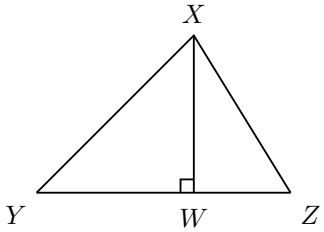


2. Two Right Triangles Sharing a Common Side

A segment from a vertex to the opposite side that meets that side at a right angle is an **altitude**, and it splits the original triangle into two right triangles that share that altitude. SOH CAH TOA applies inside each right triangle separately, and the altitude is the link that connects the two calculations.

Reading the shared side off both triangles

Complete the following work for the diagram shown.
XW is the altitude from X to base YZ, meeting it at W.



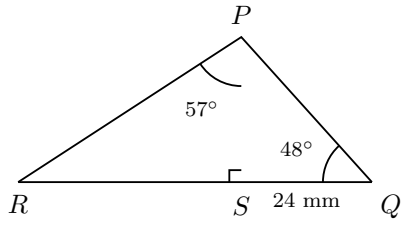
In right triangle XWY: $\tan(\angle Y) = \dfrac{XW}{\text{_____}}$, so
$YW =$ _____.

In right triangle XWZ: $\tan(\angle Z) = \dfrac{XW}{\text{_____}}$, so
$WZ =$ _____.

Since W lies between Y and Z: $YZ =$ _____ + _____.

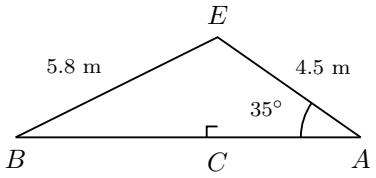
Example 2 — An altitude shared by two right triangles

In the diagram, $QS = 24$ mm, $\angle PQS = 48°$, and $\angle RPS = 57°$, where PS is perpendicular to RQ. Determine the length of QR to the nearest millimetre.



Example 3 — Finding an angle through the shared side

In the diagram, EC is perpendicular to BA, $EB = 5.8$ m, $EA = 4.5$ m, and $\angle EAC = 35°$. Determine the measure of $\angle BEC$ to the nearest degree.

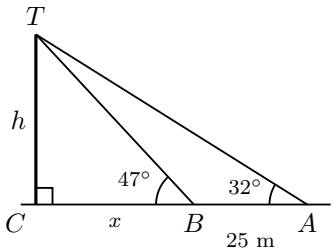


3. Two Sightings, One Unknown Distance

Sometimes *neither* right triangle has a known side: an observer measures two angles of elevation from two points on a line, and only the distance *between* the points is known. Write each horizontal distance in terms of the unknown height h, then subtract to eliminate the unknown distance.

Example 4 — Two angles of elevation, 25 m apart

From point A on level ground, the angle of elevation of the top T of a building is $32°$. Walking 25 m straight toward the building to point B, the angle of elevation becomes $47°$. Determine the height h of the building to the nearest metre.



(a) Write $x = BC$ in terms of h, using $\triangle TBC$.

(b) Write $x + 25 = AC$ in terms of h, using $\triangle TAC$.

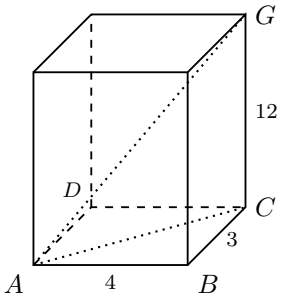
(c) Subtract (a) from (b) to eliminate x, then solve for h.

4. Extension — Extending Into Three Dimensions

A solid can also hide two right triangles that share a side. The usual plan: find a right triangle lying in the **base**, solve it, then hand its side to a right triangle standing **vertically** on it.

Example 5 — The diagonal of a box

A box has base $ABCD$ with $AB = 4$ cm and $BC = 3$ cm, and height $CG = 12$ cm (G is directly above C).



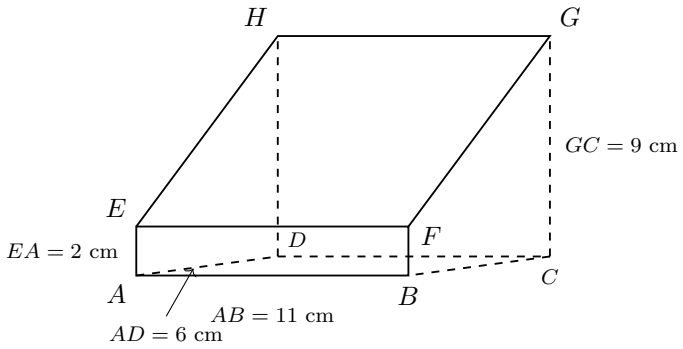
(a) Determine the length of the base diagonal AC.

(b) Determine the length of the box's diagonal AG, and the angle $\angle GAC$ it makes with the base, to the nearest degree.

The wedge below hides its right triangle in a side face instead: a trapezoid formed by two parallel vertical edges and the horizontal edge joining their bases. A horizontal line drawn across that face, from the shorter vertical edge to the taller one, creates a right triangle whose legs are the horizontal distance between the edges and the *difference* in their heights.

Example 6 — A wedge sliced from a rectangular prism

The solid below was formed by slicing a wedge off a rectangular prism, leaving a slanted top face $EFGH$. In the diagram, $HD = GC = 9$ cm, $EA = FB = 2$ cm, $AD = 6$ cm, and $AB = 11$ cm.



(a) Name four rectangles that appear as faces of this solid.

(b) Determine the measure of $\angle HEA$ to the nearest degree.

(c) Determine the length of EH to the nearest tenth of a centimetre.

Lesson 8: Trigonometry Unit Review

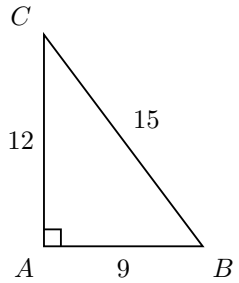
Naming sides, solving for a side, solving for an angle, elevation and depression, and splitting a figure into right triangles. (Extension: bearings and 3-D.)

1. Naming Sides and the Primary Ratios

Sine, cosine and tangent are defined only relative to a chosen acute angle of a right triangle, so every problem starts by naming which leg is **opposite** that angle, which leg is **adjacent** to it, and which side is the **hypotenuse** (always opposite the right angle, always the longest side). The two acute angles of a right triangle are complementary, which is why the sine of one always equals the cosine of the other — the **cofunction relationship**.

Example 1 — Naming sides, writing ratios, and the cofunction relationship

Triangle ABC has a right angle at A, with $AB = 9$, $AC = 12$, and $BC = 15$.



(a) Relative to angle B, name the side opposite B and the side adjacent to B (other than the hypotenuse).

opposite $B =$ _____ adjacent $B =$ _____

(b) Write $\sin B$, $\cos B$ and $\tan B$ as fractions in lowest terms.

$\sin B =$ _____ $\cos B =$ _____ $\tan B =$ _____

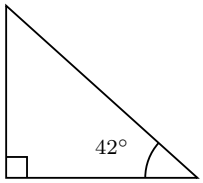
(c) Without further calculation, state $\sin C$ and $\cos C$ using the cofunction relationship. Then determine the measure of angle B itself, to the nearest degree.

2. Solving for an Unknown Side

With one acute angle and one side length known, SOH CAH TOA gives any other side: name the known side and the unknown side relative to the given angle, pick the ratio that holds both names, write the equation, then multiply or divide to isolate the unknown.

Example 2 — One angle and the adjacent leg given

In the right triangle below, one acute angle measures 42° and the side adjacent to it measures 11 cm.



(a) Determine the length of the side opposite the 42° angle, to the nearest tenth of a centimetre.

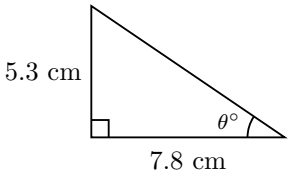
(b) Determine the length of the hypotenuse, to the nearest tenth of a centimetre.

3. Solving for an Unknown Angle

Finding an angle runs SOH CAH TOA in reverse: with two sides known, name them relative to the unknown angle, evaluate the matching ratio as a decimal, then apply $\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$ to recover the angle.

Example 3 — Two legs given

In the right triangle below, the leg opposite angle θ° measures 5.3 cm and the leg adjacent to θ° measures 7.8 cm.



(a) Determine the measure of angle θ°, to the nearest tenth of a degree.

(b) Without further calculation, state the measure of the other acute angle in the triangle.

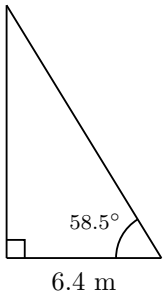
(c) Determine the length of the hypotenuse, to the nearest tenth of a centimetre.

4. Elevation, Depression, and Applied Problems

The **angle of elevation** and the **angle of depression** are both measured between a **horizontal** line of sight and the line of sight up to, or down to, an object. The horizontal at the observer and the horizontal at the object are parallel, so the angle of elevation of an object from an observer equals the angle of depression of that observer from the object. Translate the words into a labelled right triangle first; SOH CAH TOA then solves it directly.

Example 4 — A guy cable on a tower

A support cable runs from the top of a vertical broadcast tower down to an anchor point on the level ground. The cable makes an angle of 58.5° with the ground, and the anchor point is 6.4 m from the base of the tower.



(a) Determine, to the nearest tenth of a metre, how far up the tower the cable is attached.

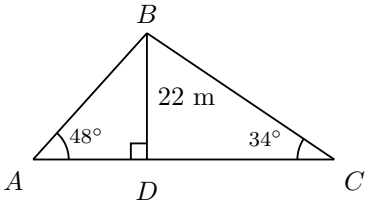
(b) Determine, to the nearest tenth of a metre, the length of the cable.

5. Combining Two Right Triangles

A triangle with no right angle can still be solved once an altitude is dropped from one vertex to the opposite side. The altitude splits the figure into two right triangles that share it as a common leg, and each of those is solved on its own; the results are then added or subtracted.

Example 5 — An altitude splitting an oblique triangle

In triangle ABC below, the altitude from B meets AC at D, with $\angle A = 48°$, $\angle C = 34°$, and $BD = 22$ m.



(a) In right triangle ABD, determine the length of AD, to the nearest tenth of a metre.

(b) In right triangle BDC, determine the length of DC, to the nearest tenth of a metre, and hence determine the length of AC.

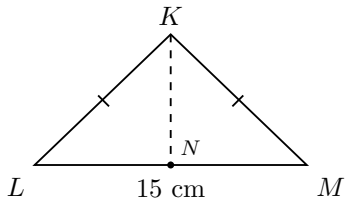
(c) State the measure of $\angle ABC$.

6. Isosceles Triangles and Area

Dropping an altitude from the apex of an isosceles triangle (where the two equal sides meet) to the base splits it into two **congruent** right triangles; the altitude bisects both the base and the apex angle. That same altitude is the height in the area formula $A = \frac{1}{2}(\text{base})(\text{height})$, so a given area supplies the height.

Example 6 — From area to sides and the apex angle

Isosceles triangle KLM has $KL = KM$ and base $LM = 15$ cm. The area of triangle KLM is 54 cm^2.



(a) Determine the height of the triangle (the distance from K to the midpoint N of LM), to the nearest tenth of a centimetre.

(b) Determine the length of the equal sides KL and KM, to the nearest tenth of a centimetre.

(c) Determine the measure of $\angle LKM$, to the nearest degree.